RAT ground stow and A32NX RAT performance

The ram air turbine (RAT) of both aircraft can be stowed again after the landing, on the ground only, as the FCOMs describe, and the A32NX RAT has the real propeller size and holds the FCOM pressure

Aircraft	A32NX and A380X
Where to find it	The aircraft systems (A32NX hydraulic RAT, A380X electrical RAT); requested from the flyPad Ground > Services page (row on `core/efb-redesign`)
Branch	`core/rat-ground-stow`, based on `master` (FlyByWire master 66021c244 merged in); independent of the other core branches. It holds the systems part (shared Rust, tests, simvar docs, CHANGELOG line). The flyPad row that sends the request is on `core/efb-redesign`, because it is a row of the Material 3 Services page: the feature works for the user only when both branches are merged. Without the flyPad row the request can be written as `L:A32NX_RAT_STOW_REQUEST` with any L:var tool; without this branch the flyPad row shows while the RAT is out but pressing it has no effect.
Status	RAT stow: tested in the sim by the user on both aircraft: OK. A32NX RAT propeller and pressure: tested in the sim by the user: OK (blue 2,558-2,567 psi). Not sim-tested on the master base; Rust tests pass there.
Pictures	No pictures: the flyPad row is one line of the Services equipment list; the A380X RAT rig and the A32NX HYD page are stock.

1. RAT stow on the ground

> **Aircraft:** A32NX and A380X | **Where:** flyPad > Ground > Services, row "RAT stow (maintenance)" | **Status:** Tested in the sim by the user: OK on both aircraft. Rust tests in systems (ram_air_turbine_deployment.rs), a320_systems (hydraulic) and a380_systems (electrical); vitest ratStow.spec on `core/efb-redesign`.

What it is for

Once the RAT had extended (both AC buses lost, RAT MAN ON, A32NX EMER ELEC PWR MAN ON), nothing stowed it again: it stayed out for the rest of the session, and on the A32NX HYD RAT FAULT stayed on. On the real aircraft the ground crew stows it after the flight.

> **NOTE**
> A320 FCOM DSC-29-10-20 : "The RAT deploys automatically if AC BUS 1 and AC BUS 2 are both lost. It can be deployed manually from the overhead panel. It can be stowed only when the aircraft is on the ground."

> **NOTE**
> A380 FCOM DSC-29-10 : "The green hydraulic system powers RAT retraction, on ground."

The systems now accept a maintenance stow request (`L:A32NX_RAT_STOW_REQUEST`) on the ground, while no extension is commanded. One shared Rust model (`ram_air_turbine_deployment.rs`) moves the strut of both aircraft, so the A320 and the A380X behave the same.

What you should see

Situation	What you should see
RAT stowed	No RAT row in the Services list.
RAT out, aircraft on the ground and stopped	Row "RAT stow (maintenance)", status "RAT extended", button **Stow**.

Stow pressed and accepted	Status "Stowing" for about 3 s; the strut moves in 1 s; the row disappears. A380X: the RAT rig retracts. A32NX: the HYD page RAT indication goes back to stowed.
RAT out in flight, or on the ground but moving	Row greyed, status "RAT extended · stow when stopped on the ground".
Stow pressed while an extension is still commanded (emergency electrical configuration, RAT MAN ON or A32NX EMER ELEC PWR MAN ON held in)	Refused: the RAT stays out.
RAT MAN ON (or the automatic extension) after a stow	The RAT extends again.
A32NX on the ground with the RAT out	HYD RAT FAULT, as the FCOM says (PRO-ABN-HYD: "The RAT is not fully stowed"); it clears after the stow.

Limitations and why

Limitation	Why	What to do
The stow needs no hydraulic pressure.	Design choice: the FCOMs say only "on the ground" (A320) and that green hydraulics power the retraction on ground (A380); the stow is a ground crew action, so it is not tied to the aircraft hydraulics being pressurised.	-
The strut travels in 1 s.	Design choice: the same speed as the stock extension; no FCOM value.	-
No RAT on the A32NX model on this branch.	The visible A32NX RAT (doors, strut, propeller) is on `core/a32nx-exterior`; with this branch alone only the HYD page, HYD RAT FAULT and the RAT pump show the stow.	Merge `core/a32nx-exterior` to see the RAT go in.
A request in flight is dropped.	Each request is evaluated once (written back to 0), so a request in flight does not stow the RAT after the landing.	Request again on the ground.
The flyPad row needs the stationary aircraft.	Design choice of the flyPad (like the other ground services); the systems themselves only need "on the ground".	Stop the aircraft.

How to use

1. After a flight with the RAT out, stop on the ground.
2. Restore normal electrical power (engines or APU generators, or external power) so that no extension is commanded; on the A32NX the RAT MAN ON and EMER ELEC PWR MAN ON pbs are not pressed.
3. flyPad > Ground > Services: press **Stow** on the row "RAT stow (maintenance)".

How to test

The flyPad row is on `core/efb-redesign`: test with both branches merged. With this branch alone, replace each "press Stow" step by setting `L:A32NX_RAT_STOW_REQUEST` to 1 with an L:var tool (SimConnect); the "row" lines then do not apply.

#	Do this	Expected result

1	A380X on the ground, engines running, parking brake set. Overhead EMER ELEC PWR panel: open the guard and press RAT MAN ON.	The RAT extends (rig out); the flyPad Services list shows "RAT stow (maintenance)", active.
2	With RAT MAN ON released, press Stow on the flyPad.	"Stowing", the rig retracts in about 1 s and the row disappears. (While RAT MAN ON is held in, the extension is still commanded and the stow is refused.)
3	A380X in flight: press RAT MAN ON, then look at the flyPad.	Row greyed ("stow when stopped on the ground"); a request written in flight is dropped.
4	Land with the RAT out, stop, press Stow.	The RAT stows.
5	A32NX on the ground, engines running. HYD panel: RAT MAN ON (or EMER ELEC PWR MAN ON).	The RAT extends: HYD page RAT out, HYD RAT FAULT (FCOM: the RAT is not fully stowed); flyPad row "RAT stow (maintenance)".
6	Press Stow on the flyPad.	The HYD page RAT goes back to stowed, HYD RAT FAULT clears, the row disappears.
7	A32NX: press RAT MAN ON again.	The RAT extends again.

2. A32NX RAT propeller size and pressure

> **Aircraft:** A32NX | **Where:** Overhead HYD panel RAT MAN ON; ECAM HYD page | **Status:** Tested in the sim by the user: OK (group 2 : blue 2,558-2,567 psi on the RAT). Before this fix the same test gave 2,680 psi. Rust tests in a320_systems (hydraulic), systems (pumps, wind_turbine).

What it is for

The A32NX RAT propeller was 1 m wide and the RAT held the blue system at about 2,680 psi. The real A320 RAT has a 29.5 in (749 mm) propeller, and the FCOM gives 2,500 psi on the RAT.

> **NOTE**
> A320 FCOM DSC-29-10-10 : "3 000 PSI (2 500 PSI when powered by the RAT)". A320 FCOM PRO-ABN-ELEC ELEC EMER CONFIG: MIN RAT SPEED 140 kt; "The RAT is capable of supplying the EMER GEN down to 125 kt, except during flare".

The propeller now has the real diameter, and its performance is retuned to the real RAT (Hamilton Sundstrand data quoted in the code): rated output from 135.5 kt, governed speed 4,800 to 6,600 rpm. The blue pump settles at about 2,500 psi on the RAT, with or without the emergency generator. The A380X RAT is not changed.

What you should see

Situation	What you should see
RAT out, 140 kt or more	Blue system about 2,500 psi (2,400 to 2,600 psi) on the ECAM HYD page.
RAT out with normal electrics (RAT MAN ON, BLUE ELEC PUMP off)	The RAT alone holds the blue system at about 2,500 psi, at 200 kt and at 140 kt.
Emergency electrical configuration, slowing down	The emergency generator keeps running down to 125 kt. Below the governed band (about 117 kt) the blue pressure fades, and the emergency generator is lost at about 110 kt.

Limitations and why

Limitation	Why	What to do

The speeds of the fade (about 117 kt) and of the generator loss (about 110 kt) are model results.	The real values are not published: the model is tuned to give the rated output from 135.5 kt and to keep the generator down to 125 kt (FCOM); below the minimum governed speed (4,800 rpm) the pump unloads, a design choice.	-

How to use

1. Nothing to set: the RAT extends as before (both AC buses lost, RAT MAN ON, or EMER ELEC PWR MAN ON).
2. Keep 140 kt or more with the RAT as the only blue source (FCOM MIN RAT SPEED).

How to test

#	Do this	Expected result
1	A32NX in flight, clean, about 5,000 ft, 200 kt, AP and A/THR ON. Overhead HYD panel: BLUE ELEC PUMP pb OFF.	The blue system has lost its electric pump.
2	Overhead HYD panel: open the guard and press RAT MAN ON. ECAM control panel: HYD key. Wait 15 s.	The RAT is out on the HYD page; blue about 2,450-2,600 psi (not 2,680).
3	FCU SPD 160 kt for 30 s; flaps 1, SPD 140 kt for 30 s; flaps 2, SPD 135 kt for 30 s.	Blue stays about 2,450-2,600 psi at every step.
4	BLUE ELEC PUMP pb back to AUTO, accelerate and retract the flaps.	Blue back to about 3,000 psi. The RAT stays out until it is stowed on the ground (see the feature above).